Parameter-Free UTR with Universal Sharp Second-Order Complexity

Abstract

We analyze a curvature-scaled universal trust-region method for unconstrained nonconvex optimization under the generalized smoothness condition $\|\nabla^3 f(x)\| \le L_0 + L_1\|\nabla f(x)\|$. The method uses exact second-order information and globally solved quadratic trust-region subproblems without knowledge of L_0 or L_1. For independently specified gradient and curvature tolerances, it attains the universal sharp four-term oracle-complexity bound, with only accuracy-independent additive logarithms. A joint scale–gradient potential controls both interior and boundary steps. We give the algorithm, the global convergence theorem, and a self-contained proof.

1 Problem setting and assumptions

We consider $\min_{x\in\mathbb{R}^n} f(x)$ under the generalized smoothness condition of **?** , Assumption 1].

Assumption 1 (Generalized smoothness)**.** *The function $f \in C^3(\mathbb{R}^n)$ is bounded below: $f_* := \inf_{x\in\mathbb{R}^n} f(x) > -\infty$. There exist constants $L_0 > 0$ and $L_1 \ge 0$ such that*

$$\|\nabla^3 f(x)\| \le L_0 + L_1\|\nabla f(x)\| \qquad \text{for every } x \in \mathbb{R}^n. \tag{1}$$

Let $\Delta := f(x_0) - f_*$. Vector norms are Euclidean and tensor norms are induced operator norms. For independent tolerances $\epsilon, \delta > 0$, an (ϵ, δ)-second-order stationary point satisfies

$$\|\nabla f(x)\| \le \epsilon, \qquad \lambda_{\min}(\nabla^2 f(x)) \ge -\delta. \tag{2}$$

Write $[z]_+ := \max\{z, 0\}$ and $\log_+ t := \max\{\log t, 0\}$ for $t > 0$, with $\log_+ 0 := 0$. The algorithm below uses quadratic trust-region subproblems and observable acceptance tests, without knowledge of L_0 or L_1.

2 Curvature-scaled parameter-free UTR

Index all trials, including rejected ones, by $k = 0, 1, \ldots$. At (x_k, σ_k), where $\sigma_k > 0$, set

$$h_k = \max\left\{\|\nabla f(x_k)\|, \frac{[-\lambda_{\min}(\nabla^2 f(x_k))]_+^2}{64\sigma_k}\right\}, \qquad R_k = \frac{\sqrt{h_k}}{4\sqrt{\sigma_k}}. \tag{3}$$

Compute a global solution s_k of

$$\min_{\|u\|\le R_k}\left\{\nabla f(x_k)^T u + \frac{1}{2}u^T\big(\nabla^2 f(x_k) + 2\sqrt{\sigma_k h_k}\, I\big)u\right\}. \tag{4}$$

The following lemma gives the classical trust-region optimality characterization of **?** , Theorem 4.1].

Lemma 1 (Trust-region optimality conditions)**.** *Suppose $R_k > 0$. A feasible vector s_k is a global minimizer of (4) if and only if there exists $\lambda_k \geq 0$ such that*

$$\begin{aligned} \big(\nabla^2 f(x_k) + (2\sqrt{\sigma_k h_k} + \lambda_k) I\big) s_k &= -\nabla f(x_k), \\ \nabla^2 f(x_k) + (2\sqrt{\sigma_k h_k} + \lambda_k) I \succeq 0, \qquad & \lambda_k(\|s_k\| - R_k) = 0. \end{aligned} \tag{5}$$

Accept the trial precisely when both inequalities hold:

$$f(x_k) - f(x_k + s_k) \geq \frac{3}{4}\sqrt{\sigma_k h_k}\,\|s_k\|^2 + \frac{1}{2}\lambda_k \|s_k\|^2, \tag{6}$$

$$\|\nabla f(x_k + s_k)\| \leq \frac{11}{16} h_k + \lambda_k \|s_k\|. \tag{7}$$

Algorithm 1: Curvature-scaled parameter-free UTR

Input: x_0, $\sigma_0 > 0$, $\epsilon > 0$, $\delta > 0$

for $k = 0, 1, \ldots$ **do**
  if $\|\nabla f(x_k)\| \leq \epsilon$ *and* $\lambda_{\min}(\nabla^2 f(x_k)) \geq -\delta$ **then**
    return x_k
  Solve (4) globally, obtaining (s_k, λ_k)
  if (6) *and* (7) *hold* **then**
    $x_{k+1} \leftarrow x_k + s_k$; retain the accepted value and gradient
    $\sigma_{k+1} \leftarrow \sigma_k / 2$
  else
    $x_{k+1} \leftarrow x_k$; $\sigma_{k+1} \leftarrow 2\sigma_k$

The stopping test precedes model construction, so $h_k > 0$ whenever a trial is formed. If $\nabla f(x_k) = 0$ and $\lambda_{\min}(\nabla^2 f(x_k)) < 0$, then

$$2\sqrt{\sigma_k h_k} = -\tfrac{1}{4}\lambda_{\min}(\nabla^2 f(x_k)).$$

The model Hessian retains a negative eigenvalue, and its global minimizer lies on the trust-region boundary. The scale has no positive lower bound, and the tolerances enter only the stopping test. Exact values and derivatives at x_k are cached and reused after a rejection; an accepted trial's value and gradient are reused at x_{k+1}.

3 Technical estimates

Fix any pair (L_0, L_1) satisfying Assumption 1. For analysis only, define

$$\begin{aligned} Q &= L_0 + L_1^{3/2}\Delta, \\ Q_0 &= \sigma_0 + L_0 + L_1\|\nabla f(x_0)\| + L_1^{3/2}\Delta, \end{aligned} \tag{8}$$

so $Q_0 \geq Q > 0$. For $\Delta > 0$, also put

$$\mathcal{L} = 1 + \log_+ \frac{Q_0}{Q} + \log_+ \frac{Q_0}{\sigma_0} + \log_+ \frac{\|\nabla f(x_0)\|}{(\Delta^2 Q)^{1/3}}. \tag{9}$$

The additive logarithmic term in the complexity bound below is $O(\mathcal{L})$ when $\Delta > 0$. All constants in the $O(\cdot)$ notation are universal numerical constants. The quantities in (8)–(9) are never used by the algorithm and do not depend on ϵ or δ.

3.1 Nonzero steps and finite backtracking

For every formed trial, write $r_k = \|s_k\|$. The index k continues to count all trials, including rejected ones.

Lemma 2 (Nonzero trial steps)**.** *Every trial formed by Algorithm 1 has $h_k > 0$, $R_k > 0$, and $r_k > 0$.*

Proof. If $h_k = 0$, both terms in its defining maximum vanish. This gives $\nabla f(x_k) = 0$ and a positive semidefinite Hessian, so the stopping test would have succeeded before the trial was formed. Thus $h_k > 0$, and $R_k > 0$ follows from $\sigma_k > 0$.

If $\nabla f(x_k) \neq 0$, the first equation in (5) rules out $s_k = 0$. If $\nabla f(x_k) = 0$, failure of the stopping test implies $\lambda_{\min}(\nabla^2 f(x_k)) < -\delta < 0$. The scale definition then gives

$$2\sqrt{\sigma_k h_k} = -\tfrac{1}{4}\lambda_{\min}(\nabla^2 f(x_k)).$$

Consequently the smallest eigenvalue of the model Hessian is $\frac{3}{4}\lambda_{\min}(\nabla^2 f(x_k)) < 0$. A sufficiently small nonzero vector in a corresponding eigenvector direction is feasible and has strictly negative model value, whereas the model value at zero is zero. Hence zero is not a global minimizer. In either case $r_k > 0$. □

Lemma 3 (Safe trial scale)**.** *Both acceptance tests hold for trial k whenever*

$$\sigma_k \geq L_0 + L_1\|\nabla f(x_k)\| + \sqrt{L_1}[-\lambda_{\min}(\nabla^2 f(x_k))]_+. \tag{10}$$

Thus the number of consecutive rejected trials at any fixed point is finite.

Proof. Taking the inner product of the first KKT equation with s_k and using Cauchy–Schwarz gives

$$\begin{aligned}(2\sqrt{\sigma_k h_k} + \lambda_k)r_k^2 &= -\nabla f(x_k)^T s_k - s_k^T \nabla^2 f(x_k) s_k \\ &\leq \|\nabla f(x_k)\| r_k + [-\lambda_{\min}(\nabla^2 f(x_k))]_+ r_k^2.\end{aligned}$$

By Lemma 2, division by r_k is legitimate. The same KKT equation identifies the linearized gradient, so

$$\begin{aligned}(2\sqrt{\sigma_k h_k} + \lambda_k)r_k &\leq \|\nabla f(x_k)\| + [-\lambda_{\min}(\nabla^2 f(x_k))]_+ r_k, \\ \|\nabla f(x_k) + \nabla^2 f(x_k)s_k\| &= (2\sqrt{\sigma_k h_k} + \lambda_k)r_k.\end{aligned} \tag{11}$$

For $u \in [0,1]$, the identity

$$\nabla f(x_k) + u\nabla^2 f(x_k)s_k = (1-u)\nabla f(x_k) + u\big(\nabla f(x_k) + \nabla^2 f(x_k)s_k\big)$$

and the triangle inequality therefore imply

$$\|\nabla f(x_k) + u\nabla^2 f(x_k)s_k\| \leq \|\nabla f(x_k)\| + [-\lambda_{\min}(\nabla^2 f(x_k))]_+ r_k. \tag{12}$$

Under (10), each nonnegative term on its right-hand side is at most σ_k. In particular,

$$\begin{aligned}L_1 h_k &= \max\left\{L_1\|\nabla f(x_k)\|, \frac{L_1[-\lambda_{\min}(\nabla^2 f(x_k))]_+^2}{64\sigma_k}\right\} \\ &\leq \max\{\sigma_k, \sigma_k/64\} = \sigma_k.\end{aligned}$$

Since $r_k \le R_k$, it follows that

$$\sqrt{L_1} r_k \le \sqrt{L_1} R_k = \frac{1}{4}\sqrt{\frac{L_1 h_k}{\sigma_k}} \le \frac{1}{4}, \qquad L_1 r_k^2 \le \frac{1}{16}. \tag{13}$$

These inequalities also hold when $L_1 = 0$ and do not divide by L_1.

Continuity of $\nabla^3 f$ on the compact trial segment ensures that

$$M_k := \max_{0 \le u \le 1} \|\nabla^3 f(x_k + u s_k)\| < \infty.$$

Taylor's integral formula for the gradient is

$$\begin{aligned}\nabla f(x_k + u s_k) = \nabla f(x_k) + u \nabla^2 f(x_k) s_k \\ + \int_0^u (u - \theta) \nabla^3 f(x_k + \theta s_k)[s_k, s_k]\, d\theta.\end{aligned}$$

Here the third derivative applied to two vectors is viewed as a vector through the Euclidean inner product. Its integral has norm at most $M_k r_k^2 \int_0^u (u-\theta)\, d\theta = \frac{1}{2} u^2 M_k r_k^2$. Combining this with (12) yields, for every $u \in [0,1]$,

$$\|\nabla f(x_k + u s_k)\| \le \|\nabla f(x_k)\| + [-\lambda_{\min}(\nabla^2 f(x_k))]_+ r_k + \tfrac{1}{2} M_k r_k^2.$$

Apply the generalized smoothness assumption at every point of the segment and then take the maximum. Using (13) and (10) gives

$$\begin{aligned}M_k &\le L_0 + L_1 \|\nabla f(x_k)\| + L_1 [-\lambda_{\min}(\nabla^2 f(x_k))]_+ r_k + \tfrac{1}{2} L_1 M_k r_k^2 \\ &\le L_0 + L_1 \|\nabla f(x_k)\| + \tfrac{1}{4}\sqrt{L_1} [-\lambda_{\min}(\nabla^2 f(x_k))]_+ + \tfrac{1}{32} M_k \\ &\le \sigma_k + \tfrac{1}{32} M_k.\end{aligned}$$

Subtracting $M_k/32$ proves $M_k \le 32\sigma_k/31$.

The scalar Taylor formula similarly gives

$$\begin{aligned}f(x_k + s_k) - f(x_k) - \nabla f(x_k)^T s_k - \tfrac{1}{2} s_k^T \nabla^2 f(x_k) s_k \\ = \frac{1}{2} \int_0^1 (1-\theta)^2 \nabla^3 f(x_k + \theta s_k)[s_k, s_k, s_k]\, d\theta.\end{aligned}$$

Its absolute value is at most $M_k r_k^3/6 \le (16/93)\sigma_k r_k^3 \le \sigma_k r_k^3$. The gradient remainder at $u = 1$ has norm at most $M_k r_k^2/2 \le (16/31)\sigma_k r_k^2 \le 3\sigma_k r_k^2$.

The first KKT equation and the positive semidefinite condition in (5) imply

$$\begin{aligned}&- \nabla f(x_k)^T s_k - \tfrac{1}{2} s_k^T \nabla^2 f(x_k) s_k \\ &= \tfrac{1}{2} s_k^T \big(\nabla^2 f(x_k) + (2\sqrt{\sigma_k h_k} + \lambda_k) I\big) s_k + \tfrac{1}{2}(2\sqrt{\sigma_k h_k} + \lambda_k) r_k^2 \\ &\ge \tfrac{1}{2}(2\sqrt{\sigma_k h_k} + \lambda_k) r_k^2.\end{aligned}$$

Since $\sigma_k r_k \le \sigma_k R_k = \sqrt{\sigma_k h_k}/4$, subtracting the scalar Taylor remainder proves

$$\begin{aligned}f(x_k) - f(x_k + s_k) &\ge \left(\sqrt{\sigma_k h_k} + \tfrac{1}{2}\lambda_k\right) r_k^2 - \sigma_k r_k^3 \\ &\ge \tfrac{3}{4}\sqrt{\sigma_k h_k}\, r_k^2 + \tfrac{1}{2}\lambda_k r_k^2.\end{aligned}$$

This is (6). For the gradient test, the remainder bound and (11) give

$$\begin{aligned}\|\nabla f(x_k+s_k)\| &\le (2\sqrt{\sigma_k h_k}+\lambda_k)r_k + 3\sigma_k r_k^2\\ &\le 2\sqrt{\sigma_k h_k}R_k + 3\sigma_k R_k^2 + \lambda_k r_k\\ &= \tfrac{1}{2}h_k + \tfrac{3}{16}h_k + \lambda_k r_k = \tfrac{11}{16}h_k + \lambda_k r_k.\end{aligned}$$

Thus (7) holds as well. During consecutive rejections the point x_k stays fixed, so the right-hand side of (10) is a fixed finite number. Each rejection doubles the positive trial scale; finitely many doublings reach that number. The resulting trial must be accepted. □

3.2 Accepted trials, descent, and step geometry

Consider any finite prefix ending immediately after its Kth accepted trial, where $K \ge 1$. Let $\tau_0 < \cdots < \tau_{K-1}$ be its accepted trial indices, and define the endpoint index $\tau_K = \tau_{K-1}+1$. This last index denotes the point after the final acceptance; it need not be an accepted trial. Rejections do not change the iterate, and therefore

$$x_{\tau_0} = x_0, \qquad x_{\tau_{j+1}} = x_{\tau_j} + s_{\tau_j} \quad (0 \le j < K). \tag{14}$$

If the prefix is extended, the endpoint index can change to the next accepted index, but its associated point remains the same through all intervening rejections.

For the scale recurrence only, use the formal convention $\sigma_{\tau_{-1}} := 2\sigma_0$; this does not introduce an additional trial. Then, for $0 \le j < K$,

$$\begin{aligned}\sigma_{\tau_j} \le \max\big\{\sigma_{\tau_{j-1}}/2,\, 2\big(L_0 + L_1\|\nabla f(x_{\tau_j})\| \\ + \sqrt{L_1}[-\lambda_{\min}(\nabla^2 f(x_{\tau_j}))]_+\big)\big\}.\end{aligned} \tag{15}$$

Indeed, the first trial at this point uses $\sigma_{\tau_{j-1}}/2$, including $j=0$ under the convention. If this trial succeeds, the first term in the maximum suffices. Otherwise the trial immediately preceding τ_j was rejected, at the same point and at scale $\sigma_{\tau_j}/2$. By Lemma 3, this rejected scale is strictly smaller than the safe threshold in (10). Multiplying that strict inequality by two proves the second alternative in (15).

For every accepted trial k, its acceptance test, $r_k > 0$, and $\sqrt{\sigma_k h_k} \ge 4\sigma_k r_k$ show

$$f(x_k) - f(x_{k+1}) \ge \tfrac{3}{4}\sqrt{\sigma_k h_k}\, r_k^2 + \tfrac{1}{2}\lambda_k r_k^2 \ge 3\sigma_k r_k^3 > 0.$$

Using (14), these inequalities telescope:

$$\begin{aligned}&f(x_{\tau_j}) - f(x_{\tau_{j+1}}) \ge 3\sigma_{\tau_j} r_{\tau_j}^3 > 0 \quad (0 \le j < K),\\ &3\sum_{j=0}^{K-1} \sigma_{\tau_j} r_{\tau_j}^3 \le \sum_{j=0}^{K-1}\big(f(x_{\tau_j}) - f(x_{\tau_{j+1}})\big)\\ &\qquad = f(x_0) - f(x_{\tau_K}) \le f(x_0) - f_* = \Delta.\end{aligned} \tag{16}$$

Each summand is nonnegative, so the same upper bound applies to any subinterval of accepted trials. No limit of iterates or infinite sum is needed for this argument.

Lemma 4 (Boundary and interior steps)**.** *Put $\rho = 11/16$. An accepted boundary trial k, for which $r_k = R_k$, satisfies*

$$\begin{aligned}\|\nabla f(x_k)\| &\le 16\sigma_k r_k^2,\\ \|\nabla f(x_{k+1})\| &\le 51\sigma_k r_k^2,\\ [-\lambda_{\min}(\nabla^2 f(x_k))]_+ &\le 32\sigma_k r_k, \qquad \lambda_k \le 10\sqrt{\sigma_k h_k}.\end{aligned} \tag{17}$$

An accepted interior trial k, for which $r_k < R_k$, satisfies

$$\begin{aligned} h_k &= \|\nabla f(x_k)\|, \qquad \lambda_k = 0, \\ \|\nabla f(x_{k+1})\| &\le \rho \|\nabla f(x_k)\|, \\ [-\lambda_{\min}(\nabla^2 f(x_k))]_+ &\le 2\sqrt{\sigma_k \|\nabla f(x_k)\|}. \end{aligned} \tag{18}$$

Proof. The definition of h_k implies both $\|\nabla f(x_k)\| \le h_k$ and $[-\lambda_{\min}(\nabla^2 f(x_k))]_+ \le 8\sqrt{\sigma_k h_k}$, without requiring a safe scale. If $r_k = R_k$, then

$$h_k = 16\sigma_k r_k^2, \qquad \sqrt{\sigma_k h_k} = 4\sigma_k r_k.$$

These identities give the first and third bounds in (17). The KKT inequality (11), which was derived before imposing (10), gives

$$\begin{aligned} \lambda_k &\le \frac{\|\nabla f(x_k)\|}{R_k} + [-\lambda_{\min}(\nabla^2 f(x_k))]_+ - 2\sqrt{\sigma_k h_k} \\ &\le 4\sqrt{\sigma_k h_k} + 8\sqrt{\sigma_k h_k} - 2\sqrt{\sigma_k h_k} = 10\sqrt{\sigma_k h_k}. \end{aligned}$$

The gradient acceptance test now yields

$$\begin{aligned} \|\nabla f(x_{k+1})\| &\le \tfrac{11}{16} h_k + \lambda_k r_k \\ &\le \tfrac{11}{16}(16\sigma_k r_k^2) + 10(4\sigma_k r_k) r_k = 51\sigma_k r_k^2. \end{aligned}$$

If $r_k < R_k$, complementarity implies $\lambda_k = 0$. The positive semidefinite condition then gives

$$[-\lambda_{\min}(\nabla^2 f(x_k))]_+ \le 2\sqrt{\sigma_k h_k}.$$

If the curvature term were strictly larger than the gradient term in the definition of h_k, it would be positive and give

$$2\sqrt{\sigma_k h_k} = \tfrac{1}{4}[-\lambda_{\min}(\nabla^2 f(x_k))]_+ > 0.$$

The two preceding displays would force a positive number to be at most one quarter of itself, which is impossible. Therefore $h_k = \|\nabla f(x_k)\|$. Substitution into the semidefinite bound proves the last inequality of (18); substitution into the gradient acceptance test, with $\lambda_k = 0$, proves its contraction inequality. □

3.3 A contracting scale–gradient potential

Fix a finite prefix containing $K \ge 1$ accepted trials and use the indices τ_j defined above. For the potential estimates, define

$$\begin{aligned} v_j &= L_1 \|\nabla f(x_{\tau_j})\| \quad (0 \le j \le K), \\ w_j &= \sqrt{L_1}[-\lambda_{\min}(\nabla^2 f(x_{\tau_j}))]_+, \\ E_j &= L_1^{3/2}\big(f(x_{\tau_j}) - f(x_{\tau_{j+1}})\big) \quad (0 \le j < K). \end{aligned} \tag{19}$$

The definition of w_j is used for $0 \le j < K$. The descent budget (16) gives

$$E_j \ge 0, \qquad \sum_{j=0}^{K-1} E_j \le L_1^{3/2}\Delta. \tag{20}$$

These weighted quantities are used only in the proof.

Lemma 5 (Two Young estimates)**.** *For every accepted-trial index $0 \le j < K$ and every $\eta > 0$,*

$$w_j \le \frac{\sigma_{\tau_j}}{16} + 256 v_j + \frac{33554432}{81} E_j, \tag{21}$$

$$v_{j+1} \le \rho v_j + \eta \sigma_{\tau_j} + \frac{122}{\sqrt{\eta}} E_j. \tag{22}$$

Consequently, with the formal predecessor convention from (15),

$$\sigma_{\tau_j} \le \frac{4}{7} \sigma_{\tau_{j-1}} + 3L_0 + 600 v_j + 10^6 E_j. \tag{23}$$

Proof. First suppose $h_{\tau_j} = \|\nabla f(x_{\tau_j})\|$. The definition of h_{τ_j} implies

$$w_j \le 8\sqrt{\sigma_{\tau_j} v_j} \le \frac{\sigma_{\tau_j}}{16} + 256 v_j.$$

The last inequality follows from the exact identity

$$\frac{\sigma_{\tau_j}}{16} + 256 v_j - 8\sqrt{\sigma_{\tau_j} v_j} = \left(\frac{\sqrt{\sigma_{\tau_j}}}{4} - 16\sqrt{v_j} \right)^2 \ge 0.$$

This proves (21) in this case, because $E_j \ge 0$.

Now suppose the curvature term in h_{τ_j} is strictly larger than $\|\nabla f(x_{\tau_j})\|$. Then

$$2\sqrt{\sigma_{\tau_j} h_{\tau_j}} = \tfrac{1}{4}[-\lambda_{\min}(\nabla^2 f(x_{\tau_j}))]_+ > 0.$$

The positive-semidefiniteness condition in (5) yields

$$\lambda_{\tau_j} \ge \tfrac{3}{4}[-\lambda_{\min}(\nabla^2 f(x_{\tau_j}))]_+ > 0.$$

Complementarity therefore forces

$$r_{\tau_j} = R_{\tau_j} = \frac{[-\lambda_{\min}(\nabla^2 f(x_{\tau_j}))]_+}{32\sigma_{\tau_j}}.$$

Multiplying the descent inequality in (16) by $L_1^{3/2}$ now gives

$$E_j \ge 3L_1^{3/2} \sigma_{\tau_j} r_{\tau_j}^3 = \frac{3 w_j^3}{32768 \sigma_{\tau_j}^2}. \tag{24}$$

For every $z \ge 0$,

$$\frac{1}{16} + \frac{1024}{27} z^3 - z = \frac{(32z-3)^2(16z+3)}{432} \ge 0.$$

Apply this identity with $z = w_j/\sigma_{\tau_j}$ and multiply by $\sigma_{\tau_j} > 0$. Using (24), we obtain

$$w_j \le \frac{\sigma_{\tau_j}}{16} + \frac{1024 w_j^3}{27 \sigma_{\tau_j}^2} \le \frac{\sigma_{\tau_j}}{16} + \frac{1024 \cdot 32768}{27 \cdot 3} E_j.$$

Since $1024 \cdot 32768/(27 \cdot 3) = 33554432/81$, adding the nonnegative term $256 v_j$ proves (21).

To prove (22), an interior accepted step satisfies $v_{j+1} \le \rho v_j$ by (18), which is already sufficient. For a boundary step, put $u = \sqrt{L_1}\, r_{\tau_j} \ge 0$. Equations (17) and (16) give

$$v_{j+1} \le 51\sigma_{\tau_j} u^2, \qquad E_j \ge 3\sigma_{\tau_j} u^3.$$

For any $\eta > 0$, if $u \le \sqrt{\eta/51}$, then $51u^2 \le \eta$. Otherwise $u > \sqrt{\eta/51}$, so

$$51u^2 < 51\sqrt{51}\,\frac{u^3}{\sqrt{\eta}} < 366\frac{u^3}{\sqrt{\eta}},$$

where $51\sqrt{51} < 366$ follows by squaring both sides. Thus in either case

$$51u^2 \le \eta + \frac{366}{\sqrt{\eta}} u^3.$$

Multiplying by σ_{τ_j} and using the energy bound yields

$$v_{j+1} \le \eta\sigma_{\tau_j} + \frac{122}{\sqrt{\eta}} E_j \le \rho v_j + \eta\sigma_{\tau_j} + \frac{122}{\sqrt{\eta}} E_j.$$

All these arguments include $L_1 = 0$; no division by L_1 is used.

Finally, apply $\max\{a, b\} \le a + b$ for nonnegative a, b to (15) and substitute (21):

$$\begin{aligned}\sigma_{\tau_j} &\le \tfrac{1}{2}\sigma_{\tau_{j-1}} + 2L_0 + 2v_j + 2w_j \\ &\le \tfrac{1}{2}\sigma_{\tau_{j-1}} + 2L_0 + 514v_j + \tfrac{1}{8}\sigma_{\tau_j} + \frac{67108864}{81} E_j.\end{aligned}$$

Move $\sigma_{\tau_j}/8$ to the left and multiply by $8/7$:

$$\sigma_{\tau_j} \le \frac{4}{7}\sigma_{\tau_{j-1}} + \frac{16}{7} L_0 + \frac{4112}{7} v_j + \frac{536870912}{567} E_j.$$

The inequalities $16/7 < 3$, $4112/7 < 600$, and $536870912/567 < 10^6$ prove (23). □

Proposition 1 (Joint potential and scale sums)**.** *Set*

$$P_j = \sigma_{\tau_{j-1}} + 10000 v_j \quad (0 \le j \le K), \qquad P_0 = 2\sigma_0 + 10000 L_1 \|\nabla f(x_0)\|, \tag{25}$$

where the formula at $j = 0$ uses the formal predecessor convention. With $C_E = 1.3 \cdot 10^9$, one has

$$P_{j+1} \le \frac{3}{4} P_j + 4L_0 + C_E E_j \qquad (0 \le j < K). \tag{26}$$

Define

$$j_\sigma = \left\lceil \frac{\log_+(P_0/Q)}{\log(4/3)} \right\rceil, \qquad C_P = C_E + 1, \qquad C_T = 3C_P + 16 + 4C_E. \tag{27}$$

Then $j_\sigma = O(1 + \log_+(Q_0/Q))$, and every existing accepted scale satisfies

$$\sigma_{\tau_j} \le C_E Q_0, \qquad \sigma_{\tau_j} \le C_P Q \quad \text{if } j \ge j_\sigma. \tag{28}$$

For integers $b \ge j_\sigma$ and $m \ge 1$ with $b + m \le K$,

$$\sum_{j=b}^{b+m-1} \sigma_{\tau_j} \le C_T\big(mL_0 + L_1^{3/2}\Delta\big). \tag{29}$$

Proof. Choose $\eta = 10^{-6}$ in (22). Since $10000\rho = 6875$ and $10000\eta = 0.01$, adding σ_{τ_j} gives

$$\begin{aligned} P_{j+1} &\le 1.01\sigma_{\tau_j} + 6875v_j + 1.22 \cdot 10^9 E_j \\ &\le \frac{101}{175}\sigma_{\tau_{j-1}} + 3.03L_0 + 7481v_j + 1.22101 \cdot 10^9 E_j \\ &\le \tfrac{3}{4}\sigma_{\tau_{j-1}} + 7500v_j + 4L_0 + C_E E_j = \tfrac{3}{4}P_j + 4L_0 + C_E E_j. \end{aligned}$$

The second line uses (23); the third uses $101/175 < 3/4$, $7481 < 7500$, $3.03 < 4$, and $1.22101 \cdot 10^9 < C_E$.

Repeated substitution in (26) gives, for $0 \le n \le K$,

$$\begin{aligned} P_n &\le (3/4)^n P_0 + 4L_0 \sum_{i=0}^{n-1} (3/4)^{n-1-i} + C_E \sum_{i=0}^{n-1} (3/4)^{n-1-i} E_i \\ &\le (3/4)^n P_0 + 16L_0 + C_E L_1^{3/2}\Delta. \end{aligned} \tag{30}$$

For $n = 0$, the sums are empty and the first inequality is equality. For $n \ge 1$, the last line uses $\sum_{i=0}^{n-1}(3/4)^i \le 4$, the nonnegativity of E_i, and (20). Since $P_0 = 2\sigma_0 + 10000L_1\|\nabla f(x_0)\|$ and $C_E \ge 10000, 16, 2$, this also shows $P_n \le C_E Q_0$.

The definition of j_σ ensures $(3/4)^n P_0 \le Q$ whenever $n \ge j_\sigma$. For such n, (30) implies

$$P_n \le Q + 16L_0 + C_E L_1^{3/2}\Delta = 17L_0 + (C_E + 1)L_1^{3/2}\Delta \le C_P Q.$$

Because $P_0 \le 10000 Q_0$ and $Q_0 \ge Q > 0$,

$$j_\sigma \le 1 + \frac{\log 10000 + \log_+(Q_0/Q)}{\log(4/3)}.$$

Also $\sigma_{\tau_j} \le P_{j+1}$ by definition and $v_{j+1} \ge 0$. These facts prove (28).

For the suffix sum, let $W = \sum_{j=b}^{b+m-1} P_{j+1}$. The exact endpoint identity

$$\sum_{j=b}^{b+m-1} P_j = P_b + W - P_{b+m}$$

and (26) give

$$W \le \tfrac{3}{4}(P_b + W - P_{b+m}) + 4mL_0 + C_E \sum_{j=b}^{b+m-1} E_j.$$

Subtract $3W/4$ and use $P_{b+m} \ge 0$ and (20). Multiplication by 4 yields

$$\begin{aligned} \sum_{j=b}^{b+m-1} \sigma_{\tau_j} \le W &\le 3P_b + 16mL_0 + 4C_E L_1^{3/2}\Delta \\ &\le 3C_P Q + 16mL_0 + 4C_E L_1^{3/2}\Delta \\ &\le (3C_P + 16)mL_0 + (3C_P + 4C_E)L_1^{3/2}\Delta \\ &\le C_T\big(mL_0 + L_1^{3/2}\Delta\big). \end{aligned}$$

The second line uses $b \ge j_\sigma$, and the third uses $Q = L_0 + L_1^{3/2}\Delta$ and $m \ge 1$. □

4 Global convergence and universal sharp complexity

In the theorem below, $\widetilde{O}(1)$ contains only additive logarithms of problem parameters and initial data, independent of ϵ and δ; for $\Delta > 0$, it is $O(\mathcal{L})$ with $\mathcal{L}$ defined in (9). If $\Delta = 0$, the initial stopping test succeeds. The same order bounds the total exact function, gradient, and Hessian evaluations. The bound holds for every pair (L_0, L_1) satisfying Assumption 1 along the same algorithmic run. Arithmetic costs of eigenvalue computations and subproblem solves are not included.

Theorem 1 (Universal sharp complexity)**.** *Under Assumption 1, Algorithm 1 finds an (ϵ, δ)-second-order stationary point within*

$$O\left(\Delta\left[\frac{\sqrt{L_0}}{\epsilon^{3/2}} + \frac{\sqrt{L_1}}{\epsilon} + \frac{L_0^2}{\delta^3} + \frac{L_1}{\delta}\right]\right) + \widetilde{O}(1) \tag{31}$$

trust-region subproblem solves.

Proof. Fix any pair (L_0, L_1) satisfying Assumption 1. If $\Delta = 0$, then x_0 is a global minimizer. For every direction d, the function $t \mapsto f(x_0 + td)$ has a minimum at $t = 0$, so $\nabla f(x_0)^T d = 0$ and $d^T \nabla^2 f(x_0) d \geq 0$. Consequently $\nabla f(x_0) = 0$ and $\nabla^2 f(x_0) \succeq 0$; the initial stopping test succeeds without solving a subproblem.

Assume henceforth that $\Delta > 0$. We use the quantities in (8)–(9) and the technical estimates above.

An accuracy-independent gradient prefix Fix any finite prefix of K accepted steps, with the endpoint convention for τ_K specified above. All estimates below concern this finite prefix; termination has not yet been assumed. Set

$$G_* = (\Delta^2 Q)^{1/3} > 0, \qquad c_r = \frac{51}{3^{2/3}}.$$

For a boundary step, (17) and (16) give

$$\begin{aligned} \|\nabla f(x_{\tau_{j+1}})\| &\leq 51 \sigma_{\tau_j} r_{\tau_j}^2 \\ &= 51 \sigma_{\tau_j}^{1/3} (\sigma_{\tau_j} r_{\tau_j}^3)^{2/3} \\ &\leq c_r \sigma_{\tau_j}^{1/3} [f(x_{\tau_j}) - f(x_{\tau_{j+1}})]^{2/3}. \end{aligned} \tag{32}$$

Every function decrease in this expression is at most Δ. The global scale bound (28) therefore shows that each boundary step produces a gradient norm no larger than $c_r (C_E Q_0)^{1/3} \Delta^{2/3}$. Interior steps multiply the current gradient norm by at most $\rho = 11/16$. Induction on j, starting from $x_{\tau_0} = x_0$, consequently gives

$$\|\nabla f(x_{\tau_j})\| \leq H_0 := \max\left\{\|\nabla f(x_0)\|, c_r (C_E Q_0)^{1/3} \Delta^{2/3}\right\}, \quad 0 \leq j \leq K. \tag{33}$$

For $j \geq j_\sigma$, the sharper bound $\sigma_{\tau_j} \leq C_P Q$ in Proposition 1 and (32) show that a boundary step produces a gradient norm no larger than

$$H := c_r C_P^{1/3} G_*.$$

Define the nonnegative integers

$$\ell := \left\lceil \frac{\log_+(H_0/H)}{\log(16/11)} \right\rceil, \qquad s := j_\sigma + \ell. \tag{34}$$

If $K < s$, this prefix already has fewer than s accepted steps. Suppose instead that $K \geq s$. If all ℓ steps with accepted indices $j_\sigma, \ldots, s-1$ are interior, repeated use of (18) gives

$$\|\nabla f(x_{\tau_s})\| \leq \rho^\ell \|\nabla f(x_{\tau_{j_\sigma}})\| \leq \rho^\ell H_0 \leq H.$$

The last inequality follows directly from the ceiling in (34); it also holds when $\ell = 0$, since then $H_0 \leq H$. If one of these steps is a boundary step, its next-point gradient norm is at most H. Every later interior step contracts this norm, and every later boundary step again bounds it by H. Thus in both cases

$$\|\nabla f(x_{\tau_s})\| \leq H. \tag{35}$$

We next bound the prefix length explicitly in terms of the analysis parameters. Since $c_r C_P^{1/3} > 1$ and $C_E < C_P$, the definitions imply

$$\frac{H_0}{H} \leq \max\left\{ \frac{\|\nabla f(x_0)\|}{G_*}, \left(\frac{Q_0}{Q}\right)^{1/3} \right\}.$$

Using $\lceil z \rceil \leq z + 1$ for $z \geq 0$ and $\log_+ \max\{a, b\} \leq \log_+ a + \log_+ b$ for $a \geq 0$, $b > 0$, we obtain

$$\ell \leq 1 + \frac{1}{\log(16/11)} \left(\log_+ \frac{\|\nabla f(x_0)\|}{G_*} + \frac{1}{3} \log_+ \frac{Q_0}{Q} \right).$$

Together with the bound for j_σ from Proposition 1, this proves

$$s = O\left(1 + \log_+ \frac{Q_0}{Q} + \log_+ \frac{\|\nabla f(x_0)\|}{G_*}\right). \tag{36}$$

In particular, s does not depend on ϵ or δ. These indices are used only in the proof; the algorithm does not compute them or restart at s.

Sums of gradient norms and negative curvature Suppose $K \geq s + m$ for some integer $m \geq 1$, and let

$$\mathcal{J} = \{s, s+1, \ldots, s+m-1\}, \qquad T = \sum_{j \in \mathcal{J}} \sigma_{\tau_j}, \qquad S = mL_0 + L_1^{3/2}\Delta.$$

Partition $\mathcal{J}$ into the boundary indices $\mathcal{J}_\partial := \{j \in \mathcal{J} : r_{\tau_j} = R_{\tau_j}\}$ and the interior indices $\mathcal{J}_\circ := \mathcal{J} \setminus \mathcal{J}_\partial$. The scale estimate (29) and the descent budget (16) imply

$$\begin{aligned} T &\leq C_T S, \\ \sum_{j \in \mathcal{J}} \sigma_{\tau_j} r_{\tau_j}^3 &\leq \frac{1}{3} \sum_{j \in \mathcal{J}} [f(x_{\tau_j}) - f(x_{\tau_{j+1}})] \leq \frac{\Delta}{3}. \end{aligned} \tag{37}$$

Also $S \geq Q$ because $m \geq 1$.

Both cases in (17)–(18) imply the single recurrence

$$\|\nabla f(x_{\tau_{j+1}})\| \leq \rho \|\nabla f(x_{\tau_j})\| + 511 \mathbf{1}_{\{j \in \mathcal{J}_\partial\}} \sigma_{\tau_j} r_{\tau_j}^2, \qquad j \in \mathcal{J}.$$

Indeed, on an interior step the second term vanishes, while on a boundary step (17) bounds the left-hand side by the second term alone. Summing this recurrence and retaining the two endpoint

terms gives

$$\begin{aligned}
&(1-\rho)\sum_{j\in\mathcal{J}}\|\nabla f(x_{\tau_j})\| \\
&\quad\le \|\nabla f(x_{\tau_s})\| - \|\nabla f(x_{\tau_{s+m}})\| + 51\sum_{j\in\mathcal{J}_\partial}\sigma_{\tau_j}r_{\tau_j}^2 \\
&\quad\le H + 51\sum_{j\in\mathcal{J}_\partial}\sigma_{\tau_j}r_{\tau_j}^2.
\end{aligned}$$

The last line uses (35) and nonnegativity of the terminal gradient norm. For the remaining sum, Hölder's inequality with exponents $3/2$ and 3 yields

$$\begin{aligned}
\sum_{j\in\mathcal{J}_\partial}\sigma_{\tau_j}r_{\tau_j}^2 &= \sum_{j\in\mathcal{J}_\partial}(\sigma_{\tau_j}r_{\tau_j}^3)^{2/3}\sigma_{\tau_j}^{1/3} \\
&\le \left(\sum_{j\in\mathcal{J}_\partial}\sigma_{\tau_j}r_{\tau_j}^3\right)^{2/3}\left(\sum_{j\in\mathcal{J}_\partial}\sigma_{\tau_j}\right)^{1/3} \\
&\le (\Delta/3)^{2/3}T^{1/3} \le 3^{-2/3}C_T^{1/3}\Delta^{2/3}S^{1/3}.
\end{aligned}$$

Empty boundary sums equal zero, so this argument also covers a suffix consisting entirely of interior steps. Since $H = c_r C_P^{1/3}\Delta^{2/3}Q^{1/3}$ and $Q \le S$, we conclude that

$$\begin{aligned}
\sum_{j\in\mathcal{J}}\|\nabla f(x_{\tau_j})\| &\le C_g\Delta^{2/3}S^{1/3}, \\
C_g &:= \frac{c_r C_P^{1/3} + 51\cdot 3^{-2/3}C_T^{1/3}}{1-\rho}.
\end{aligned} \tag{38}$$

For the curvature sum, first use the interior estimate in (18), followed by Cauchy–Schwarz:

$$\begin{aligned}
&\sum_{j\in\mathcal{J}_\circ}[-\lambda_{\min}(\nabla^2 f(x_{\tau_j}))]_+ \\
&\quad\le 2\sum_{j\in\mathcal{J}_\circ}\sqrt{\sigma_{\tau_j}\|\nabla f(x_{\tau_j})\|} \\
&\quad\le 2\left(\sum_{j\in\mathcal{J}_\circ}\sigma_{\tau_j}\right)^{1/2}\left(\sum_{j\in\mathcal{J}_\circ}\|\nabla f(x_{\tau_j})\|\right)^{1/2} \\
&\quad\le 2\sqrt{C_T C_g}\,\Delta^{1/3}S^{2/3}.
\end{aligned}$$

For boundary indices, (17) and Hölder's inequality with exponents 3 and 3/2 instead give

$$\begin{aligned}
&\sum_{j\in\mathcal{J}_\partial}[-\lambda_{\min}(\nabla^2 f(x_{\tau_j}))]_+ \\
&\quad\le 32\sum_{j\in\mathcal{J}_\partial}\sigma_{\tau_j}r_{\tau_j} \\
&\quad= 32\sum_{j\in\mathcal{J}_\partial}(\sigma_{\tau_j}r_{\tau_j}^3)^{1/3}\sigma_{\tau_j}^{2/3} \\
&\quad\le 32\left(\sum_{j\in\mathcal{J}_\partial}\sigma_{\tau_j}r_{\tau_j}^3\right)^{1/3}\left(\sum_{j\in\mathcal{J}_\partial}\sigma_{\tau_j}\right)^{2/3} \\
&\quad\le 32\cdot 3^{-1/3}C_T^{2/3}\Delta^{1/3}S^{2/3}.
\end{aligned}$$

Adding the two estimates proves

$$\sum_{j\in\mathcal{J}}[-\lambda_{\min}(\nabla^2 f(x_{\tau_j}))]_+ \le C_{\mathrm{curv}}\Delta^{1/3}S^{2/3}, \tag{39}$$

where the universal numerical constant is

$$C_{\mathrm{curv}} := 2\sqrt{C_T C_g} + 32\cdot 3^{-1/3}C_T^{2/3}.$$

In particular, the curvature estimate concerns the current accepted points. It does not require a Taylor estimate of the next-point Hessian at a trial scale that may underestimate the safe scale.

The four-term bound and finite termination At every accepted trial the stopping test has failed. Therefore, for each $j\in\mathcal{J}$, at least one of $\|\nabla f(x_{\tau_j})\|>\epsilon$ and $[-\lambda_{\min}(\nabla^2 f(x_{\tau_j}))]_+>\delta$ holds, and

$$1 < \frac{\|\nabla f(x_{\tau_j})\|}{\epsilon} + \frac{[-\lambda_{\min}(\nabla^2 f(x_{\tau_j}))]_+}{\delta}.$$

Sum over $\mathcal{J}$, divide by m, and apply (38) and (39). This gives

$$1 < \frac{C_g\Delta^{2/3}S^{1/3}}{\epsilon m} + \frac{C_{\mathrm{curv}}\Delta^{1/3}S^{2/3}}{\delta m}.$$

For nonnegative a,b and $0<p<1$, $(a+b)^p\le a^p+b^p$: when $a+b>0$, divide by $(a+b)^p$ and use $t^p\ge t$ on $0\le t\le 1$ for $t=a/(a+b)$ and $1-t=b/(a+b)$; the case $a=b=0$ is immediate. Apply this inequality with $p=1/3$ and $p=2/3$ to $S=mL_0+L_1^{3/2}\Delta$. The two resulting expressions are

$$\begin{aligned}
\frac{\Delta^{2/3}S^{1/3}}{\epsilon m} &\le \frac{\Delta^{2/3}L_0^{1/3}}{\epsilon m^{2/3}} + \frac{\sqrt{L_1}\Delta}{\epsilon m}, \\
\frac{\Delta^{1/3}S^{2/3}}{\delta m} &\le \frac{\Delta^{1/3}L_0^{2/3}}{\delta m^{1/3}} + \frac{L_1\Delta}{\delta m}.
\end{aligned}$$

With $C_* := \max\{1, C_g, C_{\mathrm{curv}}\}$ we have therefore proved

$$1 < C_*\left[\frac{\Delta^{2/3}L_0^{1/3}}{\epsilon m^{2/3}} + \frac{\sqrt{L_1}\Delta}{\epsilon m} + \frac{\Delta^{1/3}L_0^{2/3}}{\delta m^{1/3}} + \frac{L_1\Delta}{\delta m}\right]. \tag{40}$$

To make the contradiction explicit, set $q = (8C_*)^3$ and define

$$m_* := \left\lceil q \left[1 + \Delta \left(\frac{\sqrt{L_0}}{\epsilon^{3/2}} + \frac{\sqrt{L_1}}{\epsilon} + \frac{L_0^2}{\delta^3} + \frac{L_1}{\delta} \right) \right] \right\rceil . \tag{41}$$

If $m \geq m_*$, each of the four ratios of a complexity term in the brackets of (41) to m is at most q^{-1}. The four terms in brackets in (40) are respectively the 2/3 power, the first power, the 1/3 power, and the first power of these ratios. Hence the right-hand side of (40) is at most

$$C_*(q^{-2/3} + q^{-1} + q^{-1/3} + q^{-1}) \leq 4C_* q^{-1/3} = \frac{1}{2},$$

where $q \geq 1$ was used in the second inequality. This contradicts (40).

Consequently no finite accepted prefix can have $K \geq s + m_*$: if it did, its first m_* accepted steps after s would give the contradiction just established. Prefixes with $K < s$ already satisfy the same upper bound. Thus every finite accepted prefix obeys

$$\begin{aligned} K < s + m_* = O\Bigg(& \Delta \left(\frac{\sqrt{L_0}}{\epsilon^{3/2}} + \frac{\sqrt{L_1}}{\epsilon} + \frac{L_0^2}{\delta^3} + \frac{L_1}{\delta} \right) \\ & + 1 + \log_+ \frac{Q_0}{Q} + \log_+ \frac{\|\nabla f(x_0)\|}{G_*} \Bigg). \end{aligned} \tag{42}$$

The constants here are numerical and independent of all problem parameters and tolerances. An infinite sequence of trials could not contain only finitely many acceptances: after its last acceptance, or from the initial state if no step were accepted, the point would remain fixed, and Lemma 3 would force acceptance after finitely many doublings unless the algorithm had already stopped. An infinite run would therefore have arbitrarily long finite accepted prefixes, contrary to (42). The algorithm terminates in finitely many trials, and its stopping test gives the required (ϵ, δ)-second-order stationary point.

Rejected trials, oracle evaluations, and universality If the initial point passes the stopping test, no trust-region subproblem is solved and at most three initial oracle evaluations are needed. Otherwise let $K \geq 1$ and $J \geq 0$ be the total numbers of accepted and rejected trials before termination. The stopping condition depends only on the current point, which does not change on a rejection. Hence the algorithm stops immediately after its last accepted step; no rejected trials follow that step. Immediately before this last acceptance, the scale has been halved exactly $K - 1$ times and doubled exactly J times. Thus

$$\sigma_{\tau_{K-1}} = \sigma_0 \, 2^{J-(K-1)}, \qquad J = K - 1 + \log_2 \frac{\sigma_{\tau_{K-1}}}{\sigma_0}.$$

Using (28) and $Q_0 \geq \sigma_0 > 0$, we obtain

$$\begin{aligned} J &\leq K - 1 + \log_2 C_E + \frac{1}{\log 2} \log \frac{Q_0}{\sigma_0} \\ &\leq K + C \left(1 + \log_+ \frac{Q_0}{\sigma_0} \right), \end{aligned} \tag{43}$$

for a universal numerical constant C.

Every trial solves one trust-region subproblem, so the total number of subproblems is $K + J$. The initial function value, gradient, and Hessian cost at most three oracle evaluations. Each trial

evaluates one new function value and one new gradient; on acceptance these are retained, and one new Hessian is evaluated for the next stopping test or trial. On rejection the current data are reused. Therefore the total number of exact function, gradient, and Hessian evaluations, counting these three kinds together, is at most

$$3 + 2(K + J) + K = 3 + 3K + 2J.$$

Combining (42), (43), and the definition of $\mathcal{L}$ in (9), both counts are bounded by

$$C \left[\Delta \left(\frac{\sqrt{L_0}}{\epsilon^{3/2}} + \frac{\sqrt{L_1}}{\epsilon} + \frac{L_0^2}{\delta^3} + \frac{L_1}{\delta} \right) + \mathcal{L} \right]. \tag{44}$$

This also covers initial termination because $\mathcal{L} \geq 1$. It proves the stated bound, with the additive $\widetilde{O}(1)$ term equal to $O(\mathcal{L})$ and independent of ϵ and δ.

Finally, fix any pair (L_0, L_1) satisfying the smoothness assumption. Every preceding argument applies to that pair and establishes (44). The algorithm uses neither member of the pair, nor f_* or Δ; its trial subproblems and acceptance decisions depend only on its inputs and the oracle data. The same realized run thus satisfies the bound for every admissible pair separately. This is the asserted universality. All constants introduced for the potential and counting arguments are proof constants and are not algorithmic inputs. □